

AI-Assisted Portfolio Webpage Development

Mini Project Report — Design of Web-Based Systems

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1. Project Overview

Objective

The objective of this project was to create a professional online CV and portfolio website with interactive JavaScript features and live API integration, using Claude (Anthropic) as the primary AI coding assistant throughout the development process. The final site is hosted on GitHub Pages and demonstrates a complete, polished, and functional web application.

Features Implemented

- Professional HTML CV structure with hero section, about, education, experience, skills, and contact
- CSS styling with dark/light mode, responsive layout, and custom gold/navy design system
- JavaScript interactivity: welcome modal, typing animation, reading progress bar, timeline filters, collapsible sections, add-skill input, confetti on form submit, back-to-top button, copy-to-clipboard
- API integration: GitHub Repositories API (fetches and displays public repos dynamically) and Quotes API (random motivational quote with refresh button)
- Online deployment via GitHub Pages at ahmedelmorr.github.io/CV2

2. AI Tool Used

Tool Name	Version / Platform	Purpose / Task Assisted
Claude (Anthropic)	Claude Sonnet 4.6 — claude.ai web interface	Full HTML structure, CSS design system, JavaScript feature g

Notes: Claude was used conversationally — I described what I needed in plain English and it generated complete, working code. I uploaded my existing files and it merged new features into

them without breaking previous work. It also identified and fixed a bug (missing `.hidden` CSS class) when the error messages were showing incorrectly after a successful API fetch.

3. Tasks Assisted by AI

Task 1: HTML Portfolio Structure

AI Contribution: Generated the complete `index.html` including all sections (hero, about, education, experience, skills, projects, contact), semantic tags, ARIA labels, and nav links.

Modifications Made: Adjusted personal details (name, phone, email, GPA, job history) to match my actual CV content. Replaced placeholder experience entries with my real roles.

Task 2: CSS Design System

AI Contribution: Generated the full `styles.css` with a navy/gold colour palette, CSS custom properties, responsive grid layouts, dark/light mode, card components, and all animations (fadeUp, spin, pulse).

Modifications Made: No major modifications needed. Minor tweaks to font sizing on mobile breakpoints.

Task 3: JavaScript Interactivity

AI Contribution: Generated all interactive features: typing animation, welcome modal with `localStorage`, dark mode toggle, reading progress bar, timeline category filters, collapsible skills section, add-skill input, confetti animation, copy-to-clipboard for email, contact form validation, and back-to-top button.

Modifications Made: Tested each feature manually and confirmed they worked. The confetti parameters were slightly adjusted for visual preference.

Task 4: GitHub API Integration

AI Contribution: Generated an async JavaScript function using the GitHub REST API (`api.github.com/users/:username/repos`) to fetch public repositories and render them as styled cards with name, description, language, and star count. Includes loading spinner and error handling.

Modifications Made: Set `GITHUB_USERNAME` to my actual username (AhmedElmorr). Verified the API response shape matched the expected fields.

Task 5: Quotes API Integration

AI Contribution: Generated an async function fetching from `dummyjson.com/quotes/random` and displaying the quote and author in the hero section, with a refresh button.

Modifications Made: Identified a bug where both the success content and error message showed simultaneously due to a missing `.hidden` CSS class. Claude diagnosed and fixed this by appending the utility class to `styles.css`.

Task 6: ZIP Packaging & Deployment Structure

AI Contribution: Generated the correct flat ZIP structure matching submission requirements, with all files at the root level and no nested folders.

Modifications Made: Confirmed file names matched exactly what the assignment specification required.

4. Challenges Encountered

- The initial portfolio generated by Claude used its own design rather than my Assignment 2 files. I had to upload my original files and ask it to merge — it handled this well but required a clear instruction.
- The .hidden CSS utility class was missing from the original stylesheet, causing both success content and error messages to display at the same time after a successful API call. This was caught by visually inspecting the live site and then debugging with Claude.
- GitHub Pages caching meant changes were not immediately visible after pushing new files. A hard refresh (Cmd+Shift+R) and waiting ~1 minute resolved this.
- Claude's generated experience section initially contained placeholder data that did not match my actual CV — I had to provide my real details for it to personalise the content correctly.

5. Reflections on AI Usage

What worked well?

Claude excelled at generating large, structured blocks of code quickly and correctly. Tasks that would have taken hours — such as building the full CSS design system, writing all the JavaScript features, and integrating two APIs with proper error handling — were completed in minutes. The conversational interface made iteration natural: I could say "fix the error message bug" and get a targeted solution immediately.

What did not work or needed manual correction?

Personalisation required manual input — Claude could not know my actual CV details without me providing them. Generated placeholder content had to be replaced. Additionally, the first version of the portfolio ignored my existing Assignment 2 files entirely, requiring a re-prompt with the actual files uploaded.

Did AI help improve productivity?

Significantly. Estimating conservatively, AI assistance reduced development time by roughly 70–80%. Features like the confetti animation, timeline filters, and API integration would have each required substantial research and trial-and-error without AI assistance.

Did AI affect understanding of coding concepts?

Yes, positively. Reading the AI-generated code helped reinforce patterns like IIFE (Immediately Invoked Function Expressions) for encapsulation, the async/await pattern for API calls, and DOM manipulation techniques. Seeing well-structured examples is an effective learning tool.

Would you use AI coding tools in future projects?

Yes. AI tools are most valuable as a productivity accelerator for boilerplate, structure, and common patterns — freeing time to focus on logic, personalisation, and problem-solving. They are not a

replacement for understanding the code, but a powerful complement to it.

6. Ethical and Practical Considerations

Reliability and correctness: AI-generated code is not guaranteed to be correct. The hidden CSS bug demonstrated that even straightforward code can contain subtle errors that require manual testing to catch. Developers must review and test all AI-generated output rather than treating it as authoritative.

Intellectual property and academic integrity: Using AI to generate code for academic submissions raises questions about authorship. In this project, AI was used as a tool — similar to using Stack Overflow or documentation — with the student directing, reviewing, modifying, and owning the final product. The assignment explicitly permitted AI tool usage, making this ethically acceptable within the course context.

Dependency and learning: A risk of heavy AI reliance is reduced deep understanding of the code being produced. To mitigate this, I read through all generated code, asked Claude to explain unfamiliar patterns, and manually tested every feature. AI should accelerate learning, not bypass it.

7. Online Portfolio Link

<https://ahmedelmorr.github.io/CV2/>

9. Conclusion

This project demonstrated that AI-assisted development is a genuinely transformative approach to building web applications. Using Claude as an AI coding assistant, I was able to produce a professional, feature-rich portfolio website that incorporates all requirements from Assignments 1 through 3, with two live API integrations and a polished design — in a fraction of the time it would have taken working from scratch.

The key insight is that AI tools shift the developer's role from writing every line of code to directing, reviewing, testing, and refining. This requires a solid conceptual understanding of what the code should do, even when the AI handles the syntax. Future improvements to the portfolio would include adding more projects as they are completed, integrating a backend for the contact form, and exploring additional APIs such as a weather widget or news feed.